1. About Department of MCA & M.Sc. IT

PARUL University Parul University is a legitimate university established under Gujarat Private University Act 2009, after legislation passed by the Government of Gujarat on 26 th March 2015 giving University status to Parul Group of Institutes functioning under the aegis of Parul Arogya Seva Mandal Trust.

Faculty of IT & Computer Science Faculty of IT and Computer Science, Parul University has materialized as one of the prime IT education providers at global level. Various departments under Faculty of IT and Computer Science strive in preparing IT-industry ready professionals by means of various skill development courses, vocational courses, co-curricular & extra-curricular activities, industry visits and expert lectures.

MCA Department The Department of Master of Computer Application and Master of Science in Information Technology at Parul University emphasizes on building professionals in the domain of computer applications by providing necessary environment by means of facilitating suitable blend of technical and non-technical learning experience. The department cultivates students in various curricular, co-curricular and extra-curricular activities in order to produce future system analysts, system designers, system programmers, application programmers, testing professionals, system managers, project managers, researchers and other leading positions in systems/IT department. The departments offers various subjects from diversified technical/non-technical areas such as – core IT domain, management, communication skills, mathematics & logic building and rich pool of elective subjects. The department of MCA and M.Sc. (IT) focuses on project-based learning, and hence students are motivated to work on tiny hands-on projects in practical oriented subjects to get better exposure. Moreover, throughout their MCA studies, students are required to work on around 3 mini/major projects in individual/team to get enough confidence on software-development and thereby become industry-ready.

2. Project Profile

2.1 Project Definition

Project Title

Predictive Analytics for Student Performance

Project Name

Student Performance Prediction System

2.2 Project Description

Project Description

Education plays a vital role in shaping an individual's future, and evaluating student performance is an essential part of the educational process. Teachers and educational institutions continuously monitor students' academic progress to identify strengths, weaknesses, and areas requiring improvement. However, with the increasing number of students and the growing amount of academic data, manually analyzing student performance has become time-consuming and less efficient. This creates the need for an intelligent system that can analyze historical academic records and predict future student performance accurately.

The **Predictive Analytics for Student Performance** project aims to develop a web-based application that predicts the academic performance level of students using machine learning techniques. The application analyzes various factors such as attendance, previous examination marks, study hours, assignment completion, parental education level, internet access, extracurricular activities, and other academic or demographic attributes. These factors are used as input to a machine learning model to determine the student's expected performance level, such as Excellent, Good, Average, or Poor.

The proposed system uses the Logistic Regression algorithm, a supervised machine learning classification technique widely used for predicting categorical outcomes. Logistic Regression is selected because it is simple, efficient, interpretable, and performs well for classification problems involving structured educational datasets. Before training the model, the collected dataset undergoes preprocessing steps such as handling missing values, encoding categorical variables, feature selection, and normalization to improve prediction accuracy.

The application follows a client-server architecture. The frontend is developed using HTML, CSS, Chart.js, providing a responsive and user-friendly interface. The backend is developed using Python and the Flask framework, which processes user requests, communicates with the machine learning model, and interacts with the

MySQL database for storing and retrieving student records. The machine learning model is developed using Scikit-learn, while Pandas and NumPy are used for data preprocessing and analysis.

Teachers or administrators can enter student information into the system through the web interface. The trained Logistic Regression model analyzes the provided information and predicts the student's performance level. The prediction results are displayed on the dashboard along with charts, reports, and analytics that help educators understand performance trends. The system can also identify students who may require additional academic support, allowing timely intervention and personalized guidance.

This project demonstrates the practical application of predictive analytics in education by combining web development, database management, and machine learning. It enables educational institutions to make informed decisions based on data rather than assumptions. Furthermore, the project provides students with hands-on experience in developing intelligent web applications using modern technologies and machine learning algorithms. The proposed system has the potential to improve academic monitoring, reduce dropout rates, and enhance the overall quality of education through data-driven decision-making.

2.3 Existing System and Work Environment

Before developing the proposed system, it is important to study the existing methods used by educational institutions to monitor student performance. Most institutions either rely on manual record keeping or simple student management systems that store academic information but do not provide predictive insights.

Existing System : Student Management System (SMS)

Many educational institutions use Student Management Systems that maintain student information, attendance, examination results, fee details, and academic records. These systems help automate administrative tasks.

Advantages

- Centralized student database.
- Easy management of attendance and marks.
- Faster report generation.
- Improved data storage.

Limitations

- Mainly focuses on record management.
- Does not predict future academic performance.
- Limited analytical capabilities.

- No machine learning integration.

Comparison of Existing Systems

Feature	Student Management System	Proposed System
Student Data Management	Yes	Yes
Attendance Management	Yes	Yes
Marks Management	Yes	Yes
Performance Analysis	Basic	Advanced
Machine Learning Prediction	No	Yes
Performance Level Prediction	No	Yes
Dashboard	Basic	Interactive Dashboard
Graphical Reports	Basic	Advanced
Decision Support	No	Yes

Manual Working Environment

In the current manual process, teachers collect attendance records, assignment scores, and examination marks separately. Student performance is evaluated based on experience and observation, making the process subjective and time-consuming. Since there is no automated prediction mechanism, identifying students who may perform poorly is difficult until examination results are announced. This often delays academic intervention and reduces opportunities to provide timely support.

The proposed machine learning-based system addresses these limitations by automatically analyzing student data, predicting performance levels, and providing actionable insights through dashboards and reports. It enables educational institutions to make proactive decisions that improve student learning outcomes.

2.4 Problem Statement

Problem Statement

In today's educational environment, institutions generate a large amount of academic data, including attendance records, examination scores, assignment marks, classroom participation, and other student-related information. Although this data is collected regularly, many educational institutions do not effectively utilize it to predict student performance or identify students who require academic support.

Most existing systems focus only on storing and managing student information rather than analyzing it to provide meaningful insights. Teachers often evaluate student performance manually by reviewing attendance registers and examination results. This manual process is time-consuming, prone to human errors, and becomes increasingly difficult when managing a large number of students.

Traditional student management systems provide descriptive reports but lack predictive capabilities. As a result, teachers can only identify weak-performing students after examinations are completed, reducing the opportunity for timely intervention. Students experiencing academic difficulties may continue to perform poorly because warning signs are not detected early.

Another challenge is that academic performance depends on multiple factors such as attendance, previous examination scores, study habits, assignment completion, parental education, and extracurricular activities. Analyzing all these factors manually is difficult and often leads to inconsistent evaluations.

To overcome these challenges, there is a need for an intelligent system capable of analyzing historical academic data and predicting future student performance. Such a system can assist teachers in identifying students who are likely to perform poorly and enable educational institutions to take preventive measures before final examinations.

The proposed Predictive Analytics for Student Performance Using Machine Learning system addresses these issues by applying the Logistic Regression algorithm to classify students into different performance levels such as Excellent, Good, Average, and Poor. The system provides prediction results, graphical reports, and performance analytics that support informed academic decision-making.

Problems Identified

- Manual analysis of student performance is time-consuming.
- Existing systems only store data without predictive analysis.
- Difficult to identify academically weak students at an early stage.
- Large volumes of educational data remain underutilized.
- Teachers cannot easily analyze multiple performance factors simultaneously.

- Lack of intelligent decision-support systems in educational institutions.
- Delayed academic intervention affects student success.

2.5 Need for New System

Need for the Proposed System

Educational institutions require a modern system that not only stores student information but also transforms academic data into meaningful insights. With increasing student enrollment and digital record management, manually evaluating every student's academic progress has become impractical.

The proposed system fulfills this requirement by introducing predictive analytics using machine learning. Instead of waiting for final examination results, teachers can predict student performance in advance and provide personalized guidance to students who may require additional academic support.

The new system offers several advantages over traditional methods by automating data analysis, reducing human effort, and improving prediction accuracy. Machine learning enables the system to recognize relationships between various academic factors and classify students according to their expected performance levels.

The application also improves institutional decision-making by providing dashboards, statistical reports, and graphical visualizations that simplify performance monitoring. Teachers can quickly identify attendance issues, declining academic trends, and students at risk of poor performance.

Another important reason for developing this system is the growing adoption of data analytics in education. Educational institutions increasingly rely on data-driven decision-making to improve teaching quality, curriculum planning, and student success. Predictive analytics supports this objective by converting historical academic records into actionable insights.

The proposed system also reduces paperwork, minimizes calculation errors, improves data accessibility, and maintains a centralized database that can be accessed whenever required.

Reasons for Developing the New System

- Automate student performance analysis.
- Reduce manual effort and human errors.
- Predict student performance before examinations.
- Identify academically weak students at an early stage.
- Improve teacher decision-making.
- Generate analytical reports and dashboards.

- Maintain centralized student records.

2.6 Proposed System & Features

Proposed System

The proposed system is a Machine Learning-based Student Performance Prediction System designed to analyze historical academic records and predict student performance levels. The application integrates web technologies, database management, and machine learning techniques to provide an intelligent educational analytics platform.

The system allows teachers and administrators to enter student information, manage academic records, and generate performance predictions using the Logistic Regression algorithm. The prediction results are displayed through interactive dashboards and reports, helping educators identify students who may require additional academic support.

The application follows a client-server architecture where the frontend interacts with the Flask backend. The backend processes requests, communicates with the machine learning model, and retrieves or stores information in the MySQL database.

System Architecture

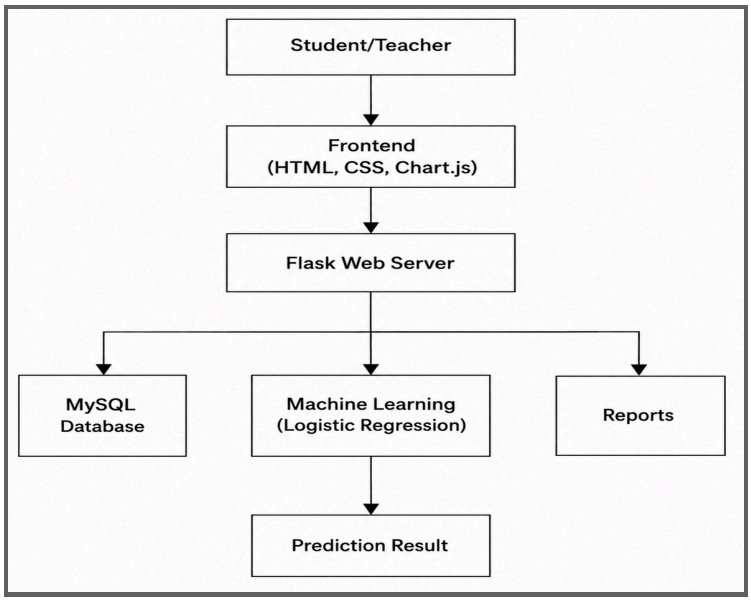


Fig. 1 System Architecture

(This diagram shows the main components and flow of the proposed system)

Modules of the Proposed System

Module 1: User Authentication

Description

This module provides secure access to the application.

Features

- Admin Login
- Teacher Login
- Password Authentication
- Session Management
- Logout

Module 2: Student Management

Description

This module manages all student-related information.

Features

- Add Student
- Update Student Details
- Delete Student
- View Student Profile
- Search Student
- Store Personal Information

Module 3: Academic Performance Management

Description

Stores academic details required for prediction.

Features

- Attendance Management
- Internal Marks
- Assignment Scores
- Previous Semester Marks
- Study Hours
- Extracurricular Activity Details
- Parent Education Details

Module 4: Dataset Management

Description

Manages the dataset used for training and prediction.

Features

- Upload CSV Dataset
- View Dataset
- Data Validation
- Remove Duplicate Records
- Data Cleaning

Module 5: Data Preprocessing

Description

Prepares raw data before model training.

Features

- Handle Missing Values
- Feature Selection
- Label Encoding
- Data Normalization
- Train-Test Split

Module 6: Machine Learning Module

Description

This is the core module of the project.

Features

- Train Logistic Regression Model
- Test Model Accuracy
- Save Trained Model (.pkl)
- Load Saved Model
- Predict Student Performance

Module 7: Prediction Module

Description

Predicts student performance level.

Features

- Accept Student Input
- Generate Prediction
- Display Performance Level
- Display Prediction Probability
- Store Prediction History

Module 8: Dashboard Module

Description

Displays analytics using charts and graphs.

Features

- Total Students
- Excellent Students
- Good Students
- Average Students

- Poor Students
- Attendance Analysis
- Marks Distribution
- Performance Comparison

Module 9: Reports Module

Description

Generates academic reports.

Features

- Student Report
- Prediction Report
- Attendance Report
- Performance Summary
- Download PDF Report
- Export Excel Report

Functionalities of the Proposed System

The proposed application provides the following functionalities:

- Secure user authentication.
- Student information management.
- Academic record management.
- Historical data storage.
- Machine learning-based prediction.
- Interactive dashboard.
- Performance analytics.
- Graphical visualization.
- Report generation.
- Database management.
- Search and filter student records.
- CSV dataset upload.
- Prediction history maintenance.

Key Features

- Machine Learning-Based Prediction
- Logistic Regression Classification
- Performance Level Prediction
- Web-Based Application
- Interactive Dashboard
- Real-Time Prediction
- Data Visualization
- Student Performance Analytics
- Report Generation
- Secure Login
- MySQL Database Integration
- Responsive User Interface
- High Accuracy Prediction
- Easy Data Management

2.7 Scope

Scope of the Project

The **Predictive Analytics for Student Performance Using Machine Learning** project is designed to help educational institutions analyze student academic data and predict future performance levels using machine learning techniques. The system can be implemented in schools, colleges, universities, coaching institutes, and other educational organizations where student academic performance needs to be monitored effectively.

The project focuses on collecting and analyzing academic information such as attendance, previous examination marks, assignment completion, study hours, parental education, internet access, and extracurricular participation. These factors are processed using the **Logistic Regression** machine learning algorithm to classify students into different performance categories such as **Excellent, Good, Average, and Poor**.

The system provides a web-based platform that enables teachers and administrators to manage student records, upload datasets, generate predictions, and visualize academic performance using interactive dashboards and reports. This allows educators to identify students who may require additional academic support before final examinations.

The application is intended primarily for educational institutions where academic data is already available. Teachers can use the system to monitor individual students as well as the overall class performance. Administrators can use

the generated reports to understand academic trends and make informed decisions regarding curriculum planning and student support programs.

Since the application is web-based, it can be accessed through any modern web browser within the institution's network or over the internet with appropriate authentication. The centralized database ensures that student information is securely stored and easily accessible whenever required.

The proposed system is suitable for:

- Schools
- Colleges
- Universities
- Coaching Institutes
- Online Learning Platforms
- Educational Research Organizations

The system can be used:

- During every academic semester
- Before internal examinations
- Before final examinations
- During academic performance reviews
- While preparing student progress reports

The intended users of the system include:

- Teachers
- Class Coordinators
- Academic Administrators
- Department Heads
- Institution Management

Although the current project uses the Logistic Regression algorithm for performance prediction, the system architecture has been designed to support additional machine learning algorithms in the future. Algorithms such as Decision Tree, Random Forest, Support Vector Machine (SVM), and Gradient Boosting can be integrated to improve prediction accuracy and compare model performance.

Future versions of the system can also incorporate mobile application support, cloud deployment, real-time analytics, personalized learning recommendations, and integration with Learning Management Systems (LMS) or Enterprise Resource Planning (ERP) software.

Overall, the scope of this project extends beyond simple student record management by introducing intelligent predictive analytics that assists educational institutions in improving academic performance through early identification of learning difficulties.

2.8 Outcomes

Expected Outcomes

After successful implementation, the proposed system is expected to deliver the following outcomes:

- Accurate prediction of student performance levels.
- Early identification of academically weak students.
- Improved academic monitoring.
- Better decision-making using data analytics.
- Reduced manual effort in performance evaluation.
- Centralized storage of student information.
- Interactive dashboards and graphical reports.
- Better communication between teachers and academic administrators.
- Improved educational planning.
- Increased student success rate through timely intervention.

Benefits of the Proposed System

For Teachers

- Easy monitoring of student progress.
- Faster performance evaluation.
- Reduced paperwork.
- Better understanding of student learning patterns.

For Students

- Early identification of academic weaknesses.
- Timely academic guidance.
- Improved learning outcomes.
- Better preparation before examinations.

For Educational Institutions

- Efficient academic management.

- Improved decision-making.
- Better utilization of academic data.
- Increased institutional performance.

2.9 Tools & Technology Used

Technologies Used

HTML5

HTML5 is used to create the structure of the web application. It provides semantic elements that improve webpage organization and browser compatibility.

Reason for Selection

- Standard web technology
- Easy to maintain
- Supported by all browsers

CSS3

CSS3 is used for designing attractive and responsive webpages.

Reason for Selection

- Responsive layouts
- Modern user interface
- Easy customization

Python

Python is the primary programming language used for backend development and machine learning implementation.

Reason for Selection

- Simple syntax
- Rich ML libraries
- Large developer community
- Easy integration with Flask

Flask

Flask is a lightweight Python web framework used to develop the web application and connect the frontend with the backend.

Reason for Selection

- Lightweight framework
- Easy routing
- Supports RESTful development
- Easy integration with machine learning models

Scikit-learn

Scikit-learn provides the implementation of the Logistic Regression algorithm.

Reason for Selection

- Reliable ML library
- Easy model training
- High prediction performance
- Built-in evaluation metrics

Pandas

Pandas is used for loading, cleaning, filtering, and analyzing datasets.

Reason for Selection

- Efficient data manipulation
- CSV handling
- Data preprocessing support

NumPy

NumPy performs mathematical and numerical operations.

Reason for Selection

- High computational performance

- Matrix operations
- Scientific computing support

MySQL

MySQL stores student records and prediction history.

Reason for Selection

- Open source
- Reliable database
- High performance
- Easy Flask integration

Visual Studio Code

Used for coding and project development.

Reason for Selection

- Lightweight IDE
- Python support
- Flask debugging
- Git integration

Jupyter Notebook

Used for experimenting with datasets and training the machine learning model.

Reason for Selection

- Interactive coding
- Easy visualization
- Ideal for ML experimentation

2.10 Project Plan

Project Development Plan (2 Months)

Task	Week
Requirement Analysis, Problem Definition, Dataset Collection	Week 1
Data Cleaning, Data Preprocessing, Feature Selection	Week 2
Frontend Development (HTML, CSS, Chart.js)	Week 3
Backend Development (Flask + MySQL Integration)	Week 4
Machine Learning Model Development (Logistic Regression)	Week 5
Model Evaluation and Accuracy Testing	Week 6
Dashboard, Reports, Prediction Module, Testing	Week 7
Final Testing, Documentation, Presentation Preparation	Week 8

Project Timeline (Gantt Chart)

Task	W1	W2	W3	W4	W5	W6	W7	W8
Requirement Analysis, Problem Definition, Dataset Collection	■							
Data Cleaning, Data Preprocessing, Feature Selection		■						
Frontend Development (HTML, CSS, Chart.js)			■					
Backend Development (Flask + MySQL Integration)				■				
Machine Learning Model Development (Logistic Regression)					■			
Model Evaluation and Accuracy Testing						■		
Dashboard, Reports, Prediction Module, Testing							■	
Final Testing, Documentation, Presentation Preparation								■

3. Requirement Analysis

The Predictive Analytics for Student Performance Using Machine Learning system is a web-based application developed to analyze student academic data and predict student performance levels using Machine Learning. The system provides separate functionalities for Teachers and Students.

Teachers can add and update student academic information, view student performance, generate performance predictions, analyze class-level performance, view AI-based insights, and generate reports. Students can view their profile, academic performance, performance analytics, predicted performance, and AI-based insights.

The system uses Logistic Regression as the machine learning classification algorithm to classify students into different performance levels such as Excellent, Good, Average, and Poor.

3.1 Feasibility Study

3.1.1 Technical Feasibility

The project is technically feasible because the required technologies can run on a standard laptop or desktop without specialized hardware. The system uses:

- Python for backend and machine learning implementation.
- Flask for developing the web application.
- Scikit-learn for implementing Logistic Regression.
- Pandas and NumPy for data preprocessing and analysis.
- MySQL for storing student and academic information.
- HTML, CSS, Bootstrap and JavaScript for the user interface.

Logistic Regression does not require a high-performance GPU. The model can be trained on a normal computer using the selected Kaggle student performance dataset.

The application can also be tested locally using a standard web browser.

Technical Feasibility Result:
The project can be successfully developed and executed using the available hardware and open-source software technologies. Therefore, the project is technically feasible.

3.1.2 Economic Feasibility

The project is economically feasible because the major development technologies are free and open source.

Resource	Cost
Python	Free
FlasK	Free
Scikit-learn	Free
Pandas	Free
NumPy	Free
MySQL	Free
Bootstrap	Free
VS Code	Free
Jupyter Notebook	Free
Kaggle Dataset	Free
Git/GitHub	Free for basic usage

No additional expensive software or specialized machine learning hardware is required.

The project can therefore be developed using an existing computer with minimal additional expenditure.

Economic Feasibility Result:

The project can be developed within a very low budget and does not require expensive licensed software or hardware. Therefore, it is economically feasible.

3.1.3 Operational Feasibility

The system is operationally feasible because it provides a simple web-based interface for both teachers and students.

Teachers can add or update student data, view performance, generate predictions, analyze dashboards, and generate reports without requiring technical knowledge of machine learning.

Students can log in and view their own profile, academic performance, prediction results, analytics, and AI insights.

The system reduces the need for manual calculations and helps teachers identify performance patterns more easily.

Operational Feasibility Result:
The system can be easily used by teachers and students in a normal educational environment and requires only basic web application knowledge. Therefore, it is operationally feasible.

3.1.4 Social Feasibility

The system is socially feasible because it is designed to assist teachers and students rather than replace teachers' academic decisions.

Teachers can use predictions and analytics to identify students who may require additional attention. Students can use their performance prediction and AI insights to understand their academic position and improve their performance.

The system provides information as decision support. Final academic decisions remain with teachers and educational authorities.

Student information should also be protected through login authentication and role-based access so that students cannot access other students' private information.

Social Feasibility Result:
The system provides useful academic support to both teachers and students and can be integrated into an educational environment without significantly changing existing teaching practices. Therefore, it is socially feasible.

3.2 Users of the System

The proposed system contains two main users/actors:

- Teacher
- Student

3.2.1 Teacher:

Role

The Teacher is responsible for maintaining student academic information and using the system to monitor and analyze student performance.

Rights

The Teacher can:

- Login to the system.
- Add student data.
- Update student data.
- View student performance.
- View performance predictions.
- View analytics dashboard.
- View AI insights.
- Generate reports.
- Reset password.
- Responsibilities

The Teacher is responsible for:

- Entering correct student information.
- Maintaining updated academic records.
- Reviewing student performance.
- Monitoring predicted performance.
- Identifying students who may need academic support.
- Using prediction results responsibly.
- Generating reports when required.

3.2.2 Student

Role

The Student uses the system to monitor their own academic information and understand their predicted performance.

Rights

The Student can:

- Login to the system.
- View their profile.
- View academic performance.
- View performance analytics.
- View performance prediction.
- View AI insights.
- Reset password.

The Student is responsible for:

- Maintaining valid login credentials.
- Viewing and monitoring academic performance.
- Understanding performance predictions.
- Using AI insights for academic improvement.
- Informing the teacher if academic information is incorrect.

3.3 Modules of the System

Module 1: Authentication and User Management

This module manages login and password-related functionality for Teachers and Students.

Features

- Teacher login.
- Student login.
- Username/password authentication.
- Role-based access.
- Forgot password.
- Reset password.

- Session management.
- Logout.

Module 2: Student Data Management

This module is primarily used by the Teacher to maintain student information.

Features

- Add student data.
- Update student data.
- Store academic information.
- Update attendance.
- Update marks.
- Maintain student records.
- Search student information.

Module 3: Academic Performance Module

This module stores and displays academic information required for performance analysis.

Features

- Attendance information.
- Previous examination marks.
- Assignment performance.
- Study-related information.
- Other selected academic attributes.
- Individual student performance details.
- Module 4: Machine Learning & Prediction Module

Module 4: Machine Learning & Prediction Module

This is the core module of the project. The system uses Logistic Regression to predict the student's performance level based on selected input features.

Features

- Load student dataset.
- Data preprocessing.
- Feature selection.

- Model training.
- Model evaluation.
- Logistic Regression classification.
- Performance prediction.
- Save trained model.
- Load trained model.
- Prediction Output

The system can classify students into performance categories such as:

- Excellent
- Good
- Average
- Poor

Module 5: Performance Analytics Module

This module analyzes student academic information and presents it through graphical and statistical representations.

Features

- Performance analysis.
- Attendance analysis.
- Marks analysis.
- Performance-level distribution.
- Student performance trends.
- Graphical visualization.
- Individual performance analytics.

Module 6: AI Insights Module

This module provides meaningful insights based on the student's academic data and prediction results.

Features

- Display predicted performance level.
- Highlight performance-related factors.
- Identify areas requiring improvement.
- Provide performance observations.

- Display academic recommendations/insights.

For example:
"The student's attendance is below the recommended level and may negatively affect overall performance." The AI Insights module should be presented as decision support, not as a replacement for the teacher's judgment.

Module 7: Dashboard Module

The dashboard provides a summarized view of student performance.

- Teacher Dashboard
- Total students.
- Performance distribution.
- Student performance statistics.
- Prediction summary.
- Analytics charts.
- AI insights.
- Student Dashboard
- Personal performance.
- Attendance.
- Performance analytics.
- Predicted performance.
- AI insights.

Module 8: Report Generation Module

This module is mainly available to Teachers.

Features

- Generate student performance reports.
- Generate prediction reports.
- Generate performance summaries.
- View class-level reports.
- Filter reports.
- Export/download reports if implemented.

3.4 Process Model

Selected Process Model: Agile Scrum

The project is planned for approximately two months, so development can be divided into small sprints. Each sprint focuses on a specific group of functionalities.

Agile is suitable because the project contains several independent modules such as authentication, student management, machine learning, prediction, analytics, AI insights, and reports. If a problem is found in one module, it can be corrected during the next sprint without affecting the entire project.

Why Agile Scrum is Suitable:

1. Short Development Duration

The project has an eight-week development period, which can be divided into multiple sprints.

2. Incremental Development

The system can be developed module by module rather than developing everything at once.

3. Easy Testing

Each completed module can be tested before moving to the next module.

4. Flexibility

Changes in features or requirements can be accommodated during development.

5. Continuous Improvement

The machine learning model, prediction interface, dashboard, and AI insights can be improved after testing.

Agile Scrum Process

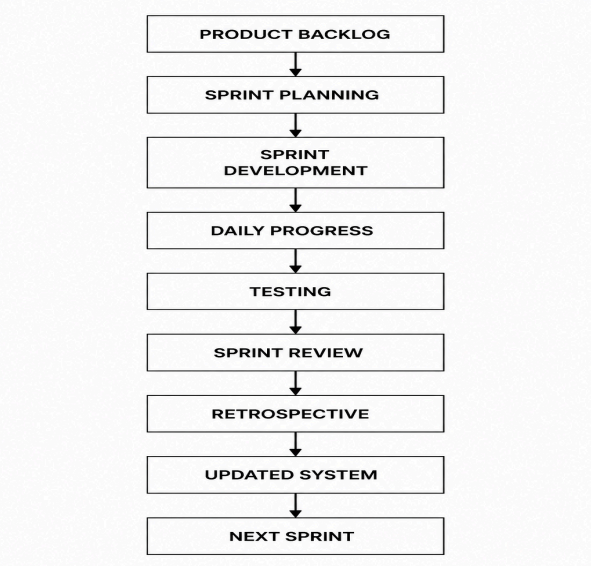


Fig. 2 Scrum Process Flow

(This diagram shows the Scrum development process followed in the project)

Scrum Sprint Plan

Sprint	Duration	Activities
Sprint 1	Week 1	Requirement Analysis, Problem Definition, Dataset Collection
Sprint 2	Week 2	Data Cleaning, Data Preprocessing, Feature Selection
Sprint 3	Week 3	Frontend Development (HTML, CSS, Chart.js)
Sprint 4	Week 4	Backend Development (Flask + MySQL Integration)
Sprint 5	Week 5	Machine Learning Model Development (Logistic Regression)
Sprint 6	Week 6	Model Evaluation and Accuracy Testing
Sprint 7	Week 7	Dashboard, Reports, Prediction Module, Testing
Sprint 8	Week 8	Final Testing, Documentation, Presentation Preparation

3.5 Hardware & Software Requirements

3.5.1 Hardware & Software Configuration at Developer's End

Hardware Requirements

Component	Configuration
Processor	Intel Core i5 / AMD Ryzen 5 or equivalent
RAM	8 GB or above
Storage	At least 20 GB free space
Display	1366 × 768 or higher
Internet	Required

Software Requirements

Software	Purpose
Windows/macOS/Linux	Operating System
Python 3.x	Backend + ML
Flask	Web framework
Scikit-learn	Machine Learning
Pandas	Data processing
NumPy	Numerical computation
MySQL	Database
VS Code	Development
Jupyter Notebook	ML experimentation

Git/GitHub	Version control
Chrome/Firefox/Safari	Application testing

3.5.2 Minimum Hardware & Software Requirements at Client/User End

Since Teachers and Students access the system through a web browser, they do not need to install Python, Flask, or machine learning libraries.

Hardware Requirements

Component	Minimum Requirement
Processor	Intel Core i3 or equivalent
RAM	4 GB
Storage	2–5 GB free
Display	1280 × 720
Network	Internet/LAN

Software Requirements

Software	Requirement
Operating System	Windows/macOS/Linux
Browser	Chrome, Firefox, Edge or Safari
Internet	Required for online deployment
Additional ML Software	Not required

4. Use Cases

A use case represents a functionality performed by an actor while interacting with the system.

The proposed system has two actors:

- Teacher
- Student

4.1 Teacher Use Cases

UC-T01: Login

Description: The Teacher enters valid credentials to access the system.

Input: Username and password

Output: Teacher dashboard

UC-T02: Add/Update Student Data

Description: The Teacher can add new student information and update existing student records.

Input: Student details and academic information

Output: Updated student record

UC-T03: View Performance Prediction

Description: The Teacher can view the performance level predicted by the machine learning model.

Input: Student academic data

Output: Predicted performance level

UC-T04: View Analytics Dashboard

Description:The Teacher can view overall student performance through charts and statistics.

Output: Analytics dashboard

UC-T05: View Student Performance

Description: The Teacher can view individual student academic performance.

Output: Student performance details

UC-T06: View AI Insights

Description: The Teacher can view system-generated insights based on student academic data and prediction results.

Output: AI-based performance insights

UC-T07: Generate Reports

Description: The Teacher can generate student or class-level performance reports.

Output: Performance report

UC-T08: Reset/Forgot Password

Description: The Teacher can recover or reset the account password.

Output: Updated password

4.2 Student Use Cases

UC-S01: Login

Description: The Student enters valid credentials to access the system.

Output: Student dashboard

UC-S02: View Profile

Description: The Student can view their personal profile information.

Output: Student profile

UC-S03: View Academic Performance

Description: The Student can view their academic information such as marks and attendance.

Output: Academic performance information

UC-S04: View Performance Analytics

Description: The Student can view graphical/statistical information about their academic performance.

Output: Performance analytics

UC-S05: View Performance Prediction

Description: The Student can view the performance level predicted by the machine learning model.

Output: Excellent / Good / Average / Poor

UC-S06: View AI Insights

Description: The Student can view insights generated from their academic information and predicted performance.

Output: AI insights and improvement areas

UC-S07: Reset/Forgot Password

Description: The Student can recover or reset their account password.

Output: Updated password

5. Use Case Diagram

Your diagrams are already structured well for the two-actor system. For the final report, keep two separate diagrams, exactly like the images you provided.

5.1 Consolidated Use Case Diagram

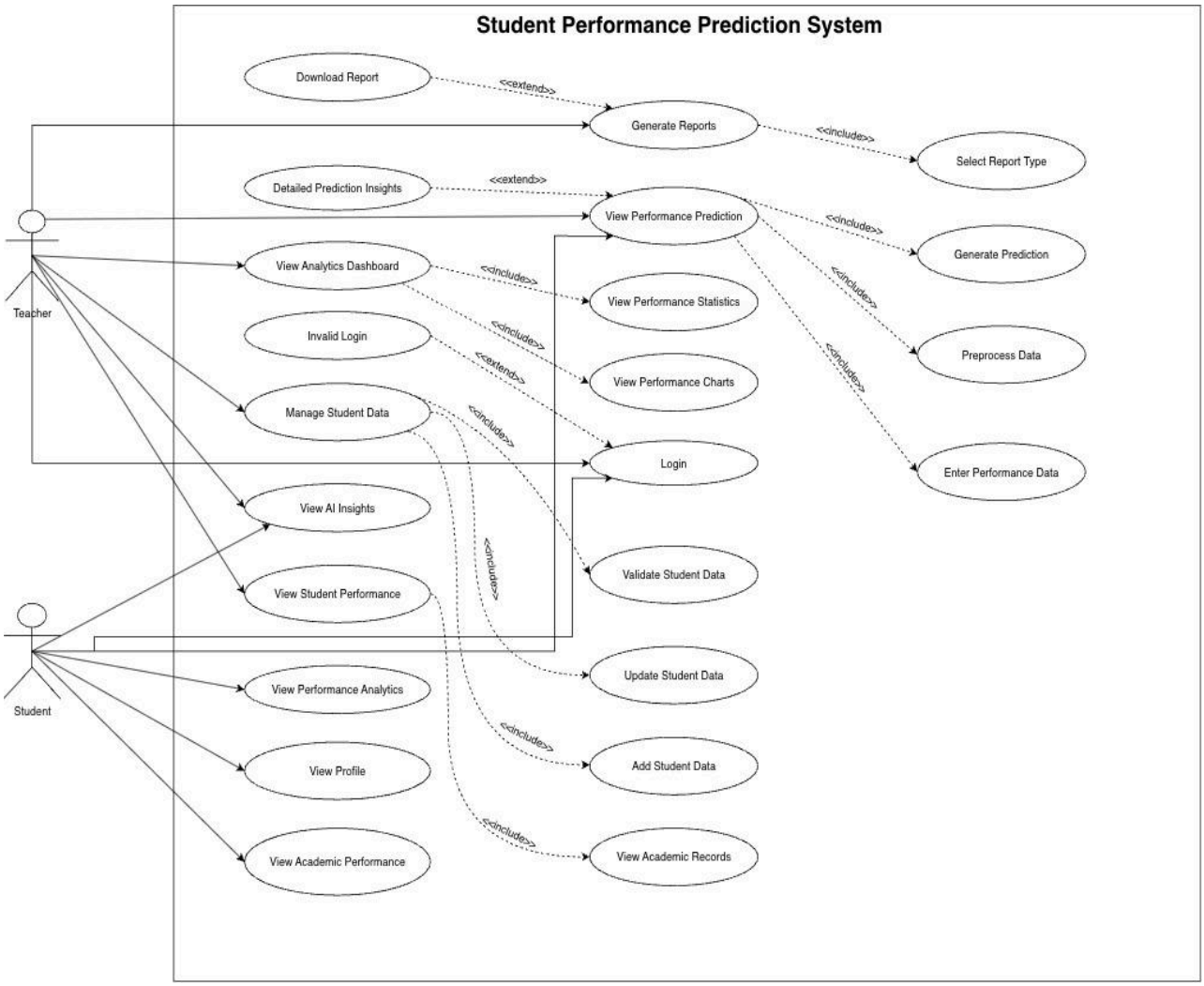


Fig. 3 Consolidated Use Case Diagram

(This diagram depicts the overall interaction of the actors with the Predictive Student Performance System)

5.2 Teacher Use Case Diagram

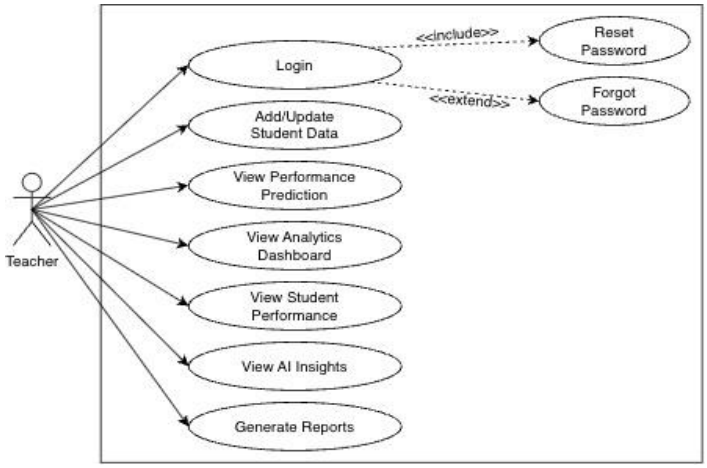


Fig. 4 Teacher Use Case Diagram

(This diagram shows the interaction of the Teacher with the system)

5.3 Student Use Case Diagram

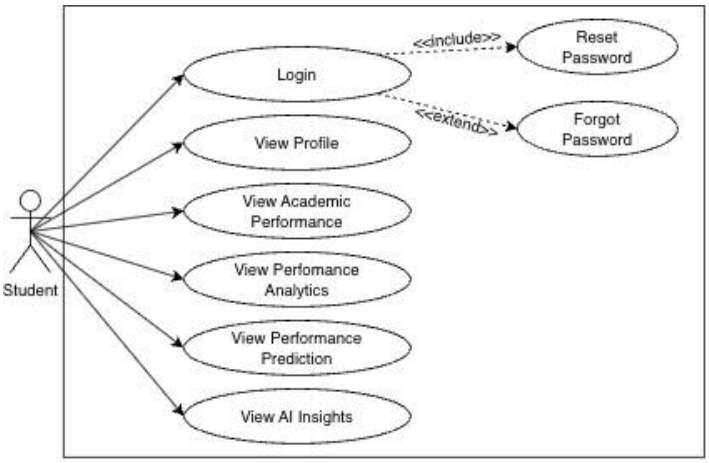


Fig. 5 Student Use Case Diagram

(This diagram shows the interaction of the Student with the system)